

Relations between VWC and potential predictors:

Marginal significance ($0.05 > p < 0.01$) will be accepted as criterion to include the predictor in the following FFS predictor selection as the present analyses should only be a pre selection to identify predictors where no relationship could be assumed at all.

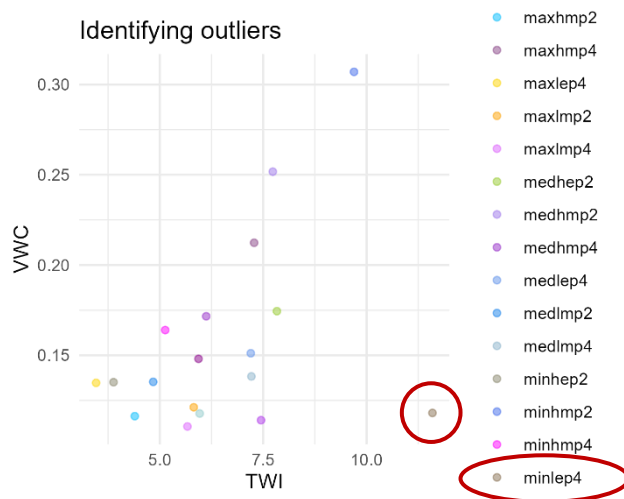
Static

Mean VWC (mean per site and for the whole measuring period) to extracted TWI

There was an outlier which was removed for the final model, outlier identified using this plot:

Note: without removing the outlier the VWC~TWI relation was not significant

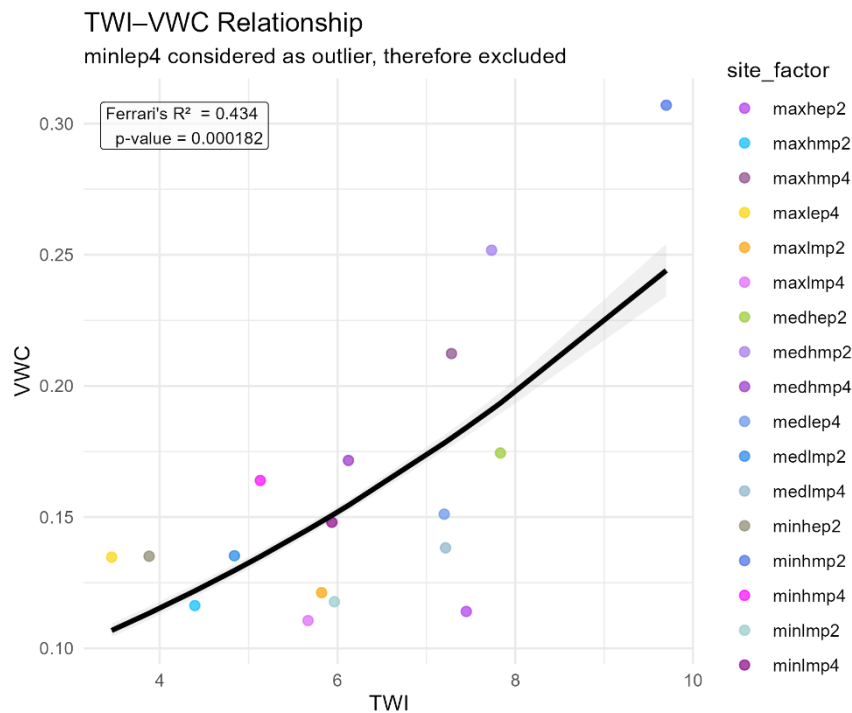
The “outlier” is one of the/ the driest Plots



Model:

```
plot_TWI_ex <- glmmTMB(moist_mean ~ TWI_mean, data = data_TWI_ex, family = beta_family())
```

Final results, Konfidenzband in grau



Slope and Aspect

Slope and Aspect turned out to be not significant.

Dynamic predictors

Site individual VWC ~ temperature sensor models

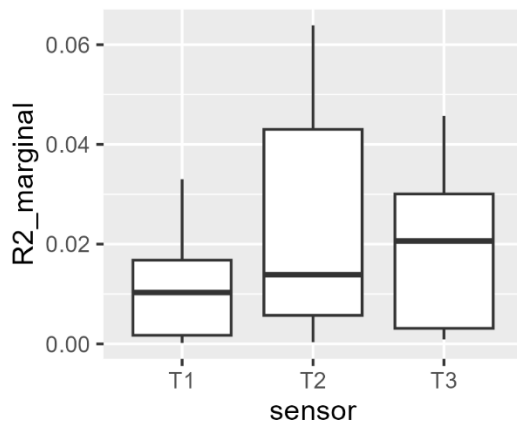
Temperature T1 (soil)- T3 (12cm above ground)

The boxplots show Nakagawa's R^2 for significant ($p < 0.05$) per site relationships between VWC~Temperature. Week of the year was considered as random effect in each model.

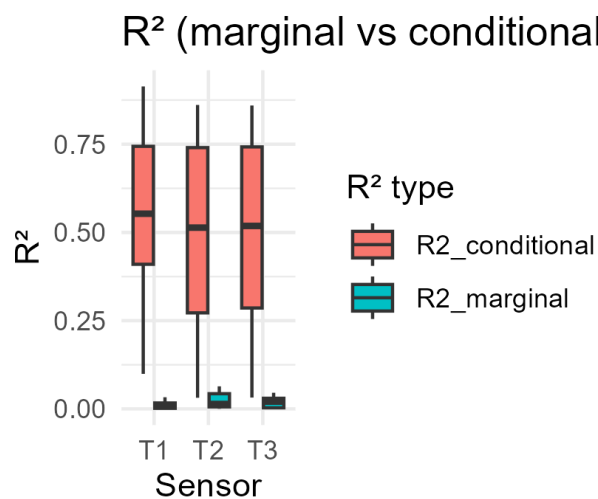
[model equation: model = VWC~ temperature sensor + (1|year_week), data = df, family = beta_family()]].

For different sensors, different sites were not significant. Number of significant sites per sensor (total number of sites = 18):

- T1: 15
- T2: 16
- T3: 14



The boxplots below shows, that, for all sites, the conditonal R2, (here the year_week as random effect) explains a lot more of the variance than the marginal R2 (fixed effect of temperature sensor). Trying to include year_week as interaction leads to convergence problems, most likely because of an issues with the degrees of freedom).



Overall T1 models, site_id as random effect (interaction leads to convergence problem)

All significant (***) BUT: Nakagawa's R2 looks like this: most of the variance explained by conditional R2, because it's just very little was is explained by R2, results are displayed as table not as plot (no meaningful trendline)

Temperature sensor	Marginal R2	Conditional R2
T1	3.979916e-05	0.74091
T2	0.0006223981	0.7414311
T2	0.0002686013	0.7408205

Because of this pattern, we calculated a model for the rolling sd for each timestep for the previous 24h to the measuring time. This sd was first smoothed and then scaled for computation time reasons.

Model equation:

```

mod_sd_T3 <- glmmTMB(
  VWC_moisture ~ sd_T3_24h_scaled + (1|year_week),
  data = data_beta_wide_filtered,
  family = beta_family(),
  dispformula = ~ ar1(locality_factor + 0 | datetime),
  control = glmmTMBControl(
    profile = FALSE,
    optCtrl = list(iter.max = 20000, eval.max = 20000)
  )
)

```

All influences from the predictor variable (1model per sensor T1-T3) were significant (***).
Results for Nakagawa's R2 are displayed in the table below

predictor	Marginal R2	Conditional R2	p-value*
T1_sd_24h_4h_smoothed_scaled	0.07073217	0.2499642	< 2.2e-16 ***
T2_sd_24h_4h_smoothed_scaled	0.03065406	0.1555367	< 2.2e-16 ***
T2_sd_24h_4h_smoothed_scaled	0.005768829	0.1020301	< 2.2e-16 ***

*p-value derived from Type II Wald chisquare test

While the models are still not explaining a lot of the variance, they do explain more than the models with just the bare temperature values as predictors.

Could it be interesting to recalculate these models site wise? Could take very long I guess...
whole model needs about 2-3h + 10-20 min for r2...

Did the same thing for a 12h moving window sd, rest analog to above

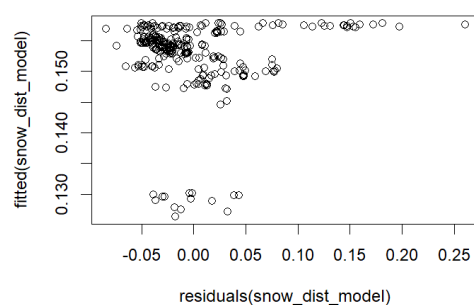
Predictor	Marginal R2	Conditional R2	p-value*
T1_sd_12h_4h_smoothed_scaled	0.02708354	0.1237812	< 2.2e-16 ***
T2_sd_12h_4h_smoothed_scaled	0.004744732	0.09105229	< 2.2e-16 ***
T2_sd_12h_4h_smoothed_scaled	0.001569611	0.09090627	< 2.2e-16 ***

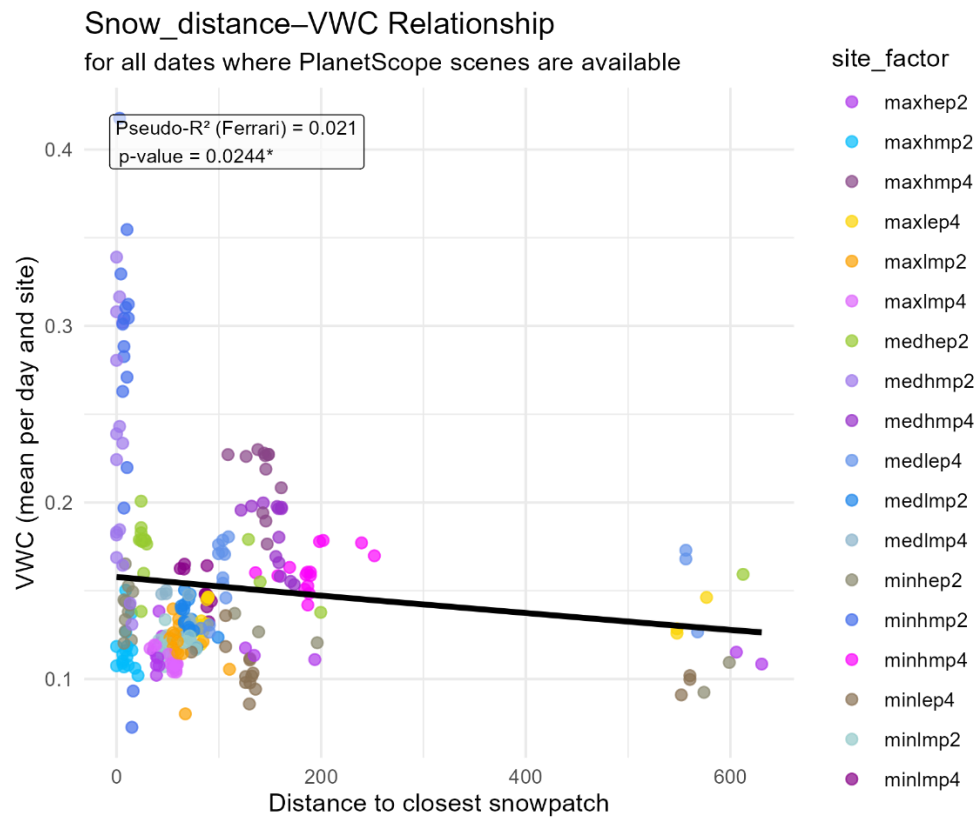
Distance to closest snowpatch

```

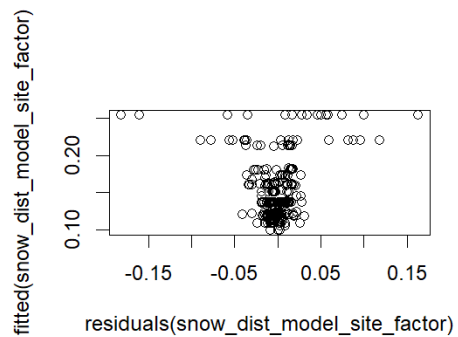
snow_dist_model <- glmmTMB(VWC_daily ~ snow_dist, data = snow_dist_beta,
  family = beta_family)

```

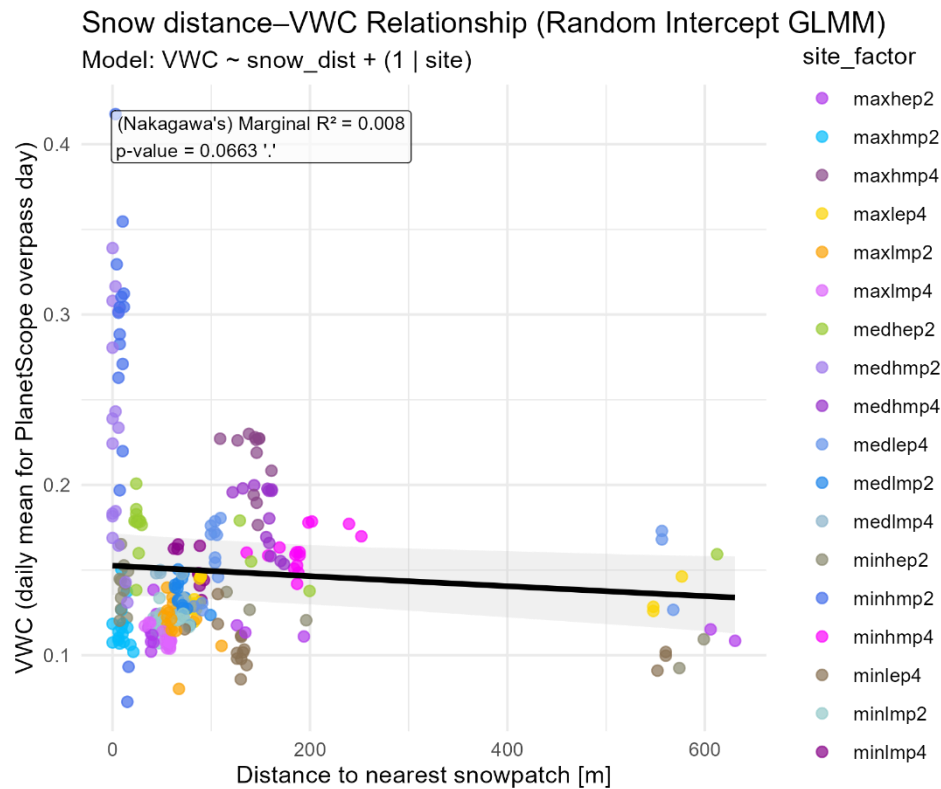




When site_id is included as random effect:



still okay



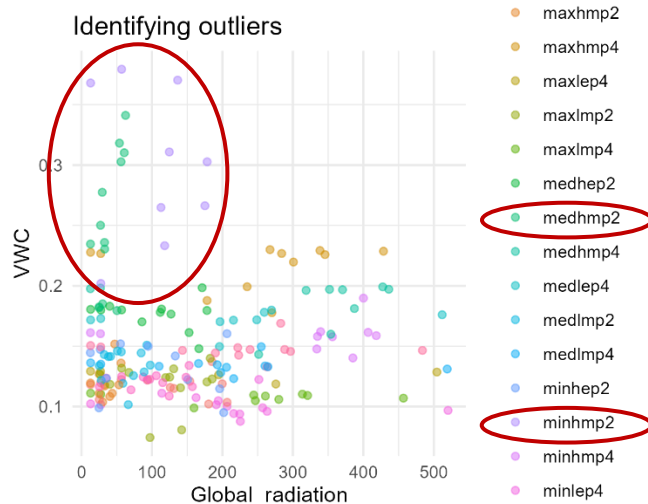
Potential solar radiation (global radiation for mean of PlanetScope overpass)

```
glob_rad_model <- glmmTMB(VWC_ps ~ glob_rad, data = glob_rad_beta,
  family = beta_family)
```

Not significant.

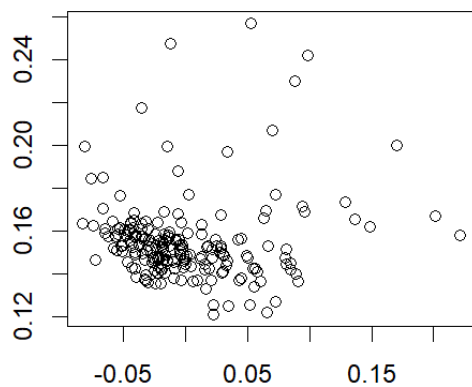
- Next step for this one; global radiation for the whole day
- Trotzdem nicht komplett verwerfen- es braucht zwar zusätzlich ein snowpatch, aber dann würden wir da ja trotzdem rein logisch betrachtet einen Zusammenhang erwarten

Again outliers?



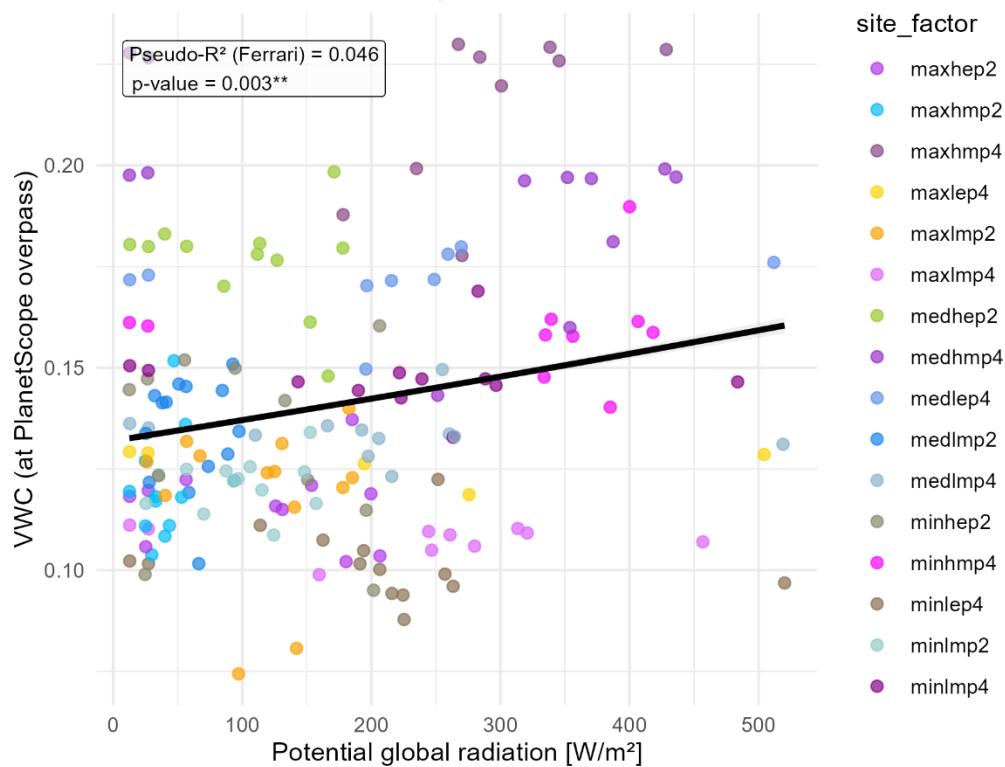
These two are the wettest spots- so there might be a different relationship for these two plots;

Model without the two above named plots



Residuals looking good, no obvious pattern

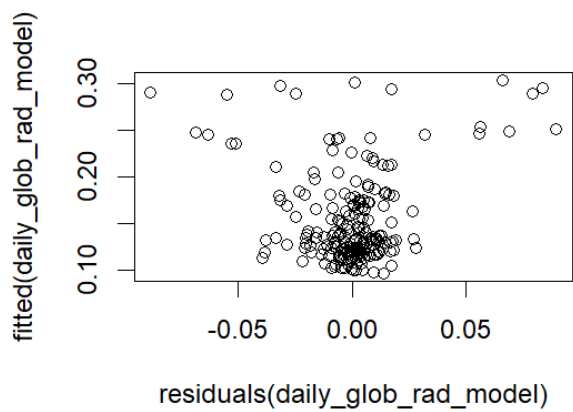
Potential global radiation–VWC Relationship
for all dates where PlanetScope scenes are available



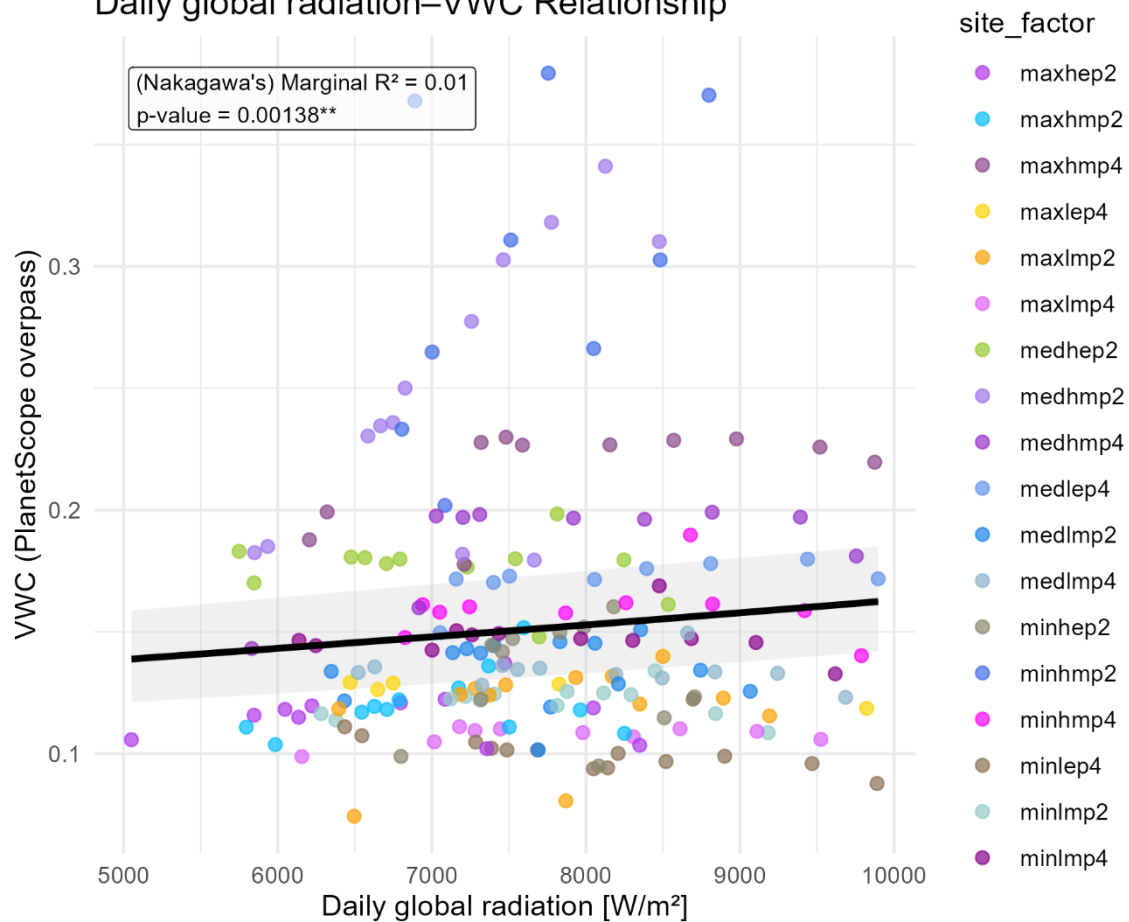
Allerdings; wenn man das mit random effect rechnet ist es nicht mehr signifikant und auch nur für einige Einzelzeitpunkte- hier statt global radiation evtl ein anderer Strahlungstyp?

Daily potential global radiation mit random effect

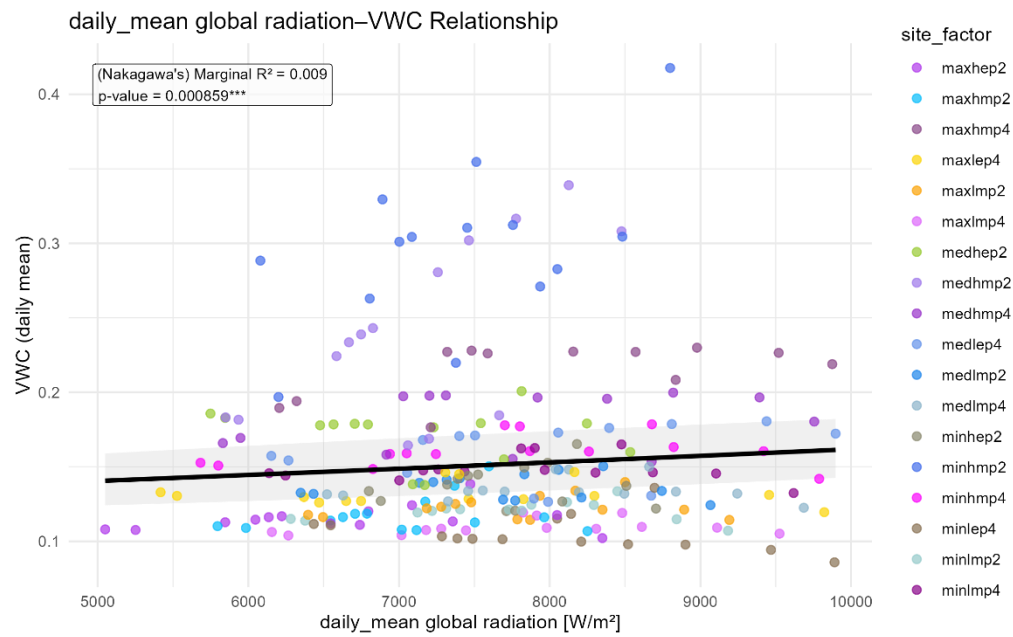
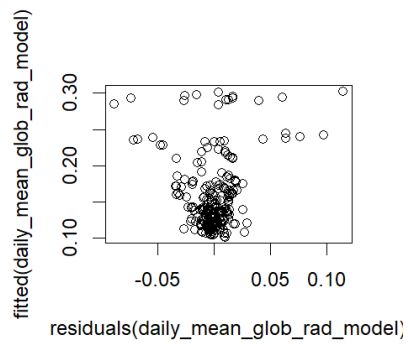
PlanetScope Overpass Zeitpunkt



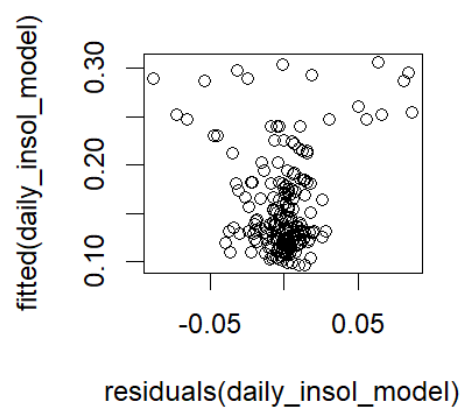
Daily global radiation–VWC Relationship



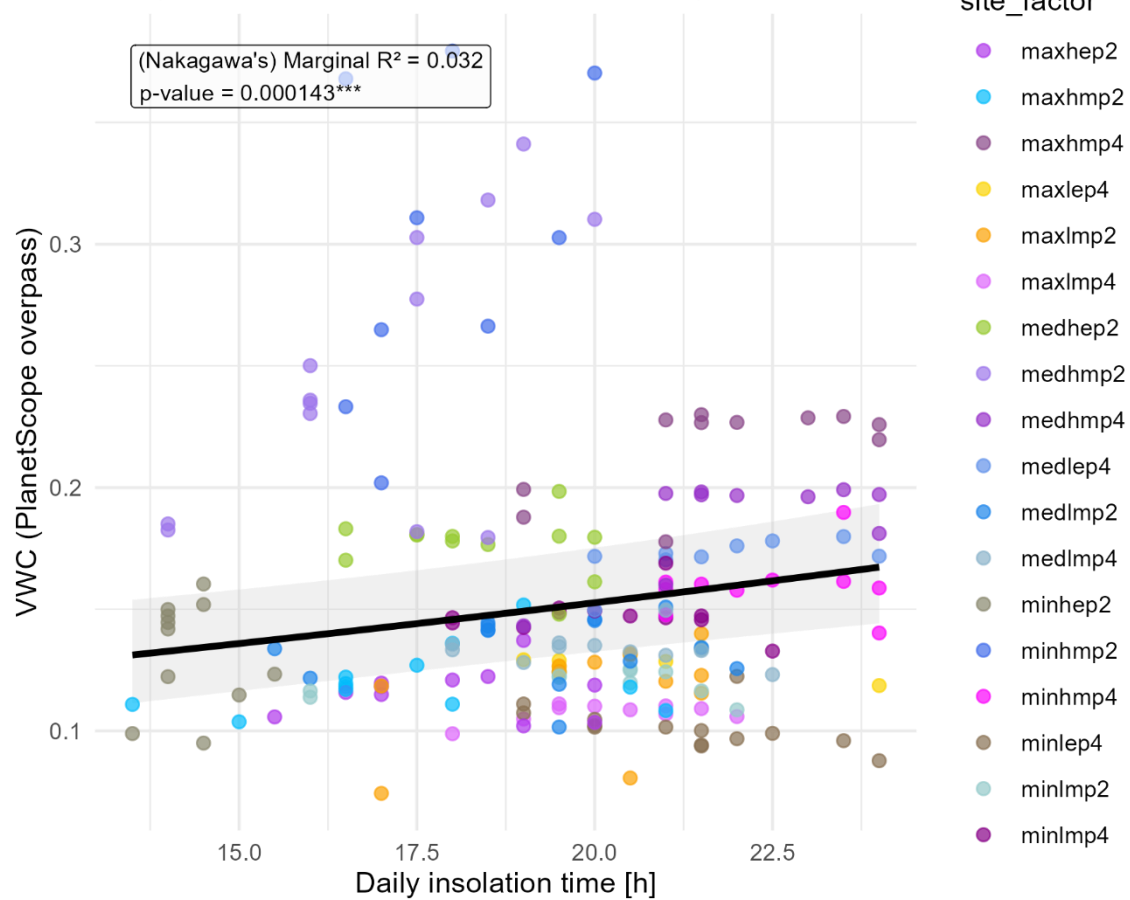
Daily VWC mean for day of Planet Scope Overpass



Daily insolation time

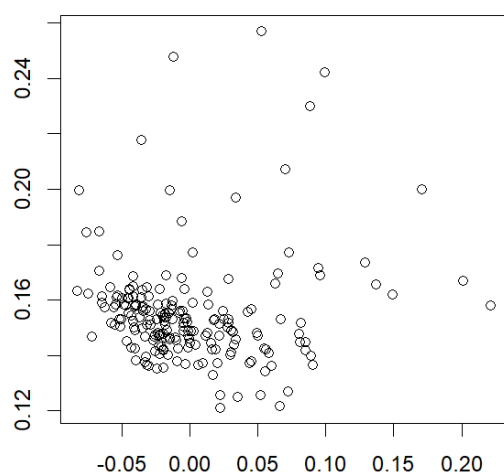


Daily insolation time–VWC Relationship



Brightness Index (compared to the overpass time)

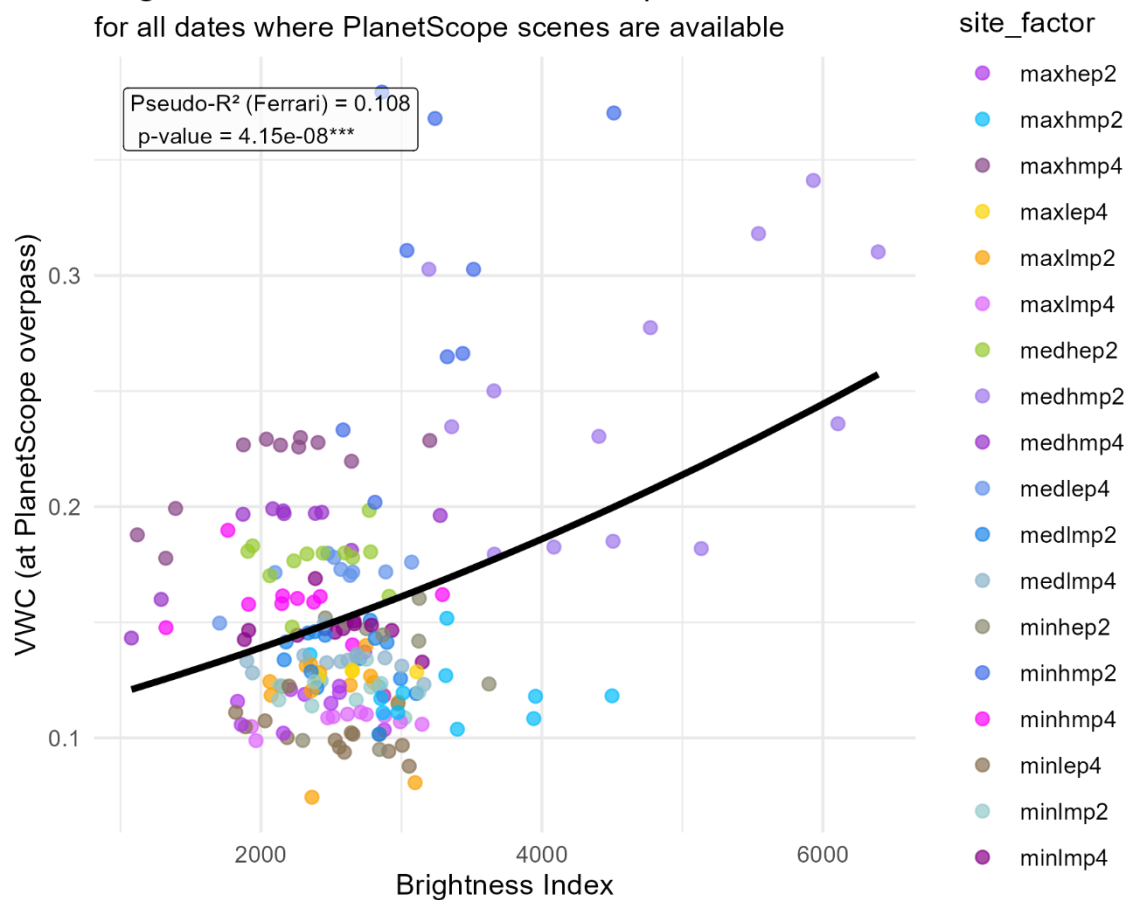
```
BI_model <- glmmTMB(VWC_ps ~ BI, data = BI_beta,
  family = beta_family)
```



No obvious pattern

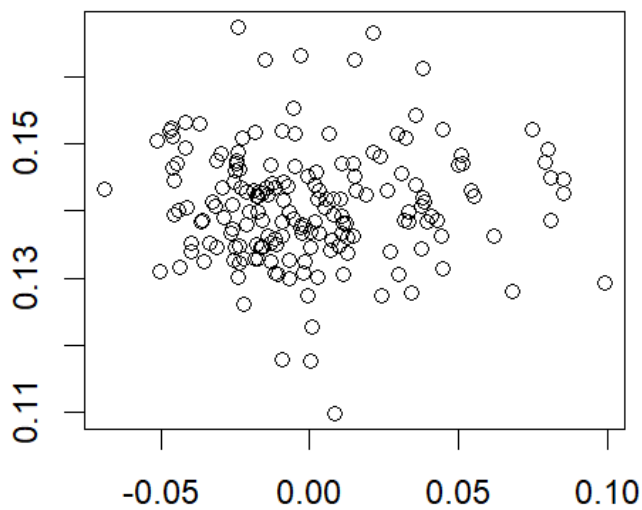
Brightness Index–VWC Relationship

for all dates where PlanetScope scenes are available



This is a bit contra intuitive- because we would assume that darker soil is wetter; BUT: The two wettest spots are the ones with the highest BI- these spots are wet because of the presence of snow (melt), indicated by the pictures taken in the field. Therefore, this makes sense. Definitely try again without these spots and see if we find a significance for lower BI indicating higher VWC. If we really have these two different relationships, this should be not a problem for the ML algorithm, because predictors like snowpatch/ distance to closest snowpatch will be included.

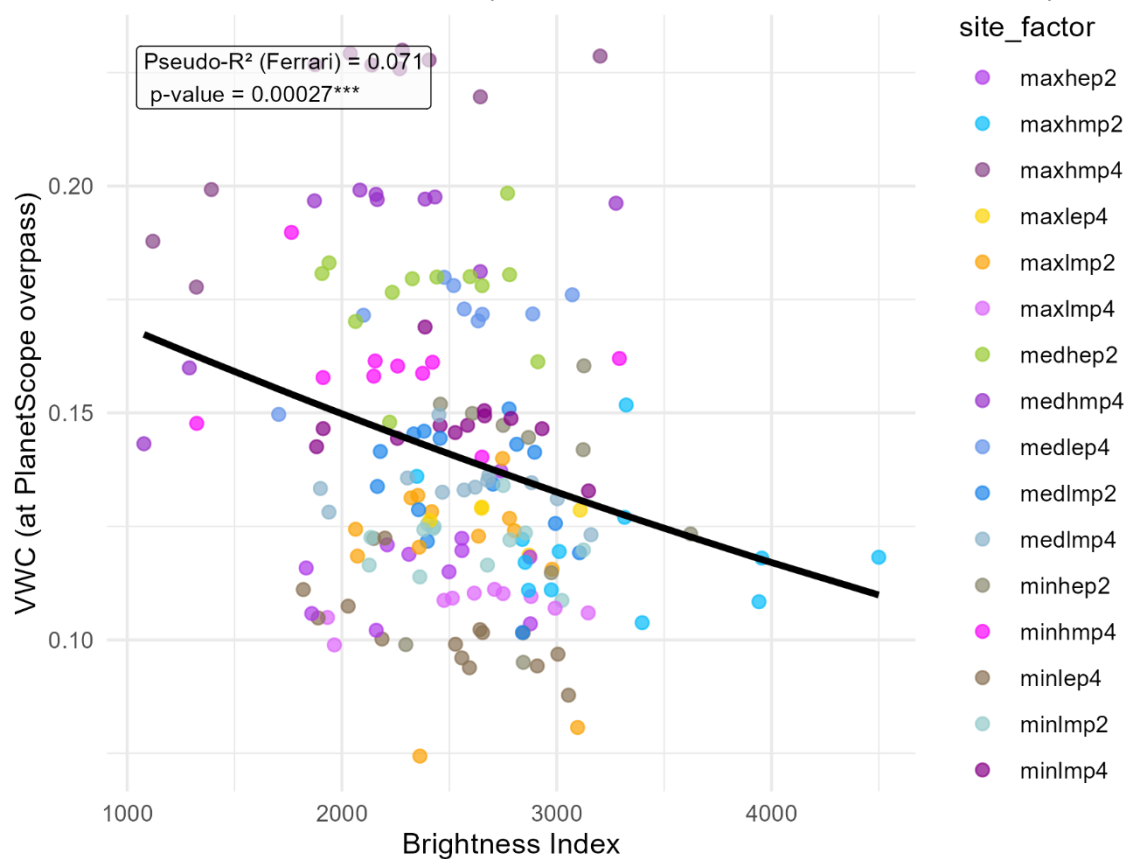
Results for the model without the two wettest spots



Residuals looking good, no obvious pattern

Brightness Index–VWC Relationship

for all dates where PlanetScope scenes are available, excl.med-/ minhmp2



Cool! This is how I expected it to look like. But the question is; should site_id be included as random effect or not?

If site_id as random effect is included its not significant anymore and if single dates are looked at not all, but most of them are significant (but I'm not to sure what that means...)

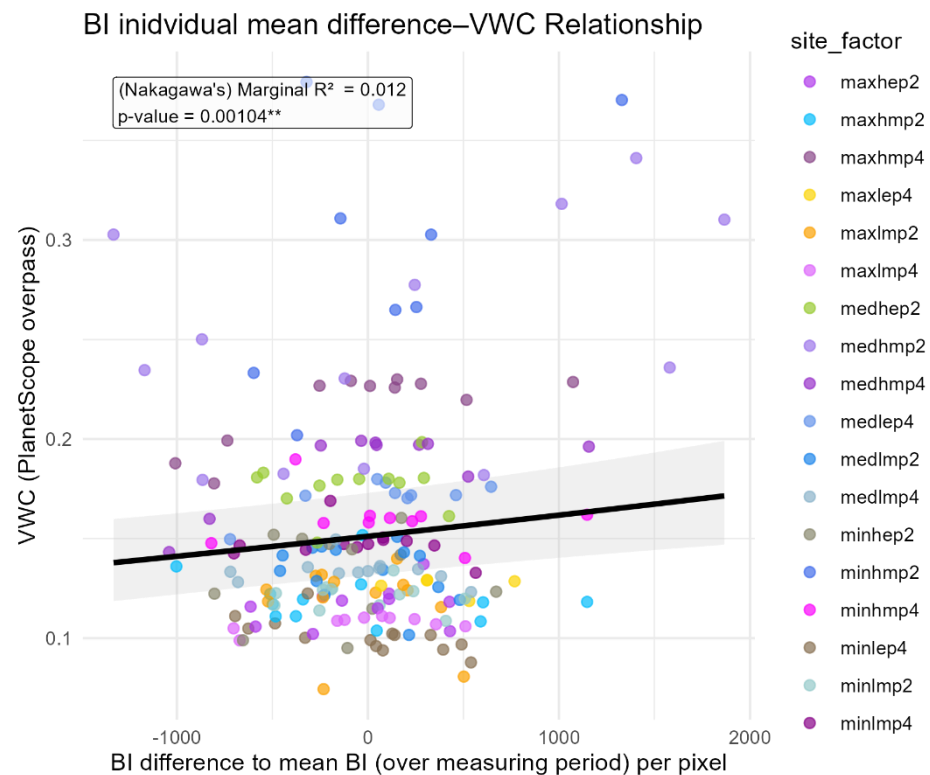
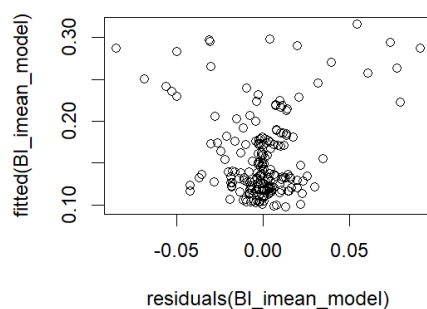
jetzt hab ich das gesamtmodell ohne random effect (re) - ist signifikant. Mit nicht mehr. Aber für alle zeitpunkte einzeln gerechnet sind wieder die meisten signifikant. Ich kann damit also zwischen den sites aber nicht je site im zeitlichen verlauf unterscheiden? aber was, wenn sich VWC für einige sites im zeitlichen verlauf einfach nicht stark ändert? Ich würde sagen, das Ergebnis zeigt trotzdem, dass das ein brauchbarer Prädiktor sein kann.

Brightness Index (BI) difference to mean BI for measuring period and individual pixel

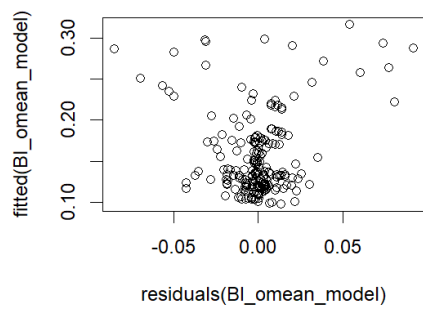
BI_mean_raster = terra::mean(BI_stack, na.rm = T)

BI_individual_mean_rasterstack = BI_stack - BI_mean_raster

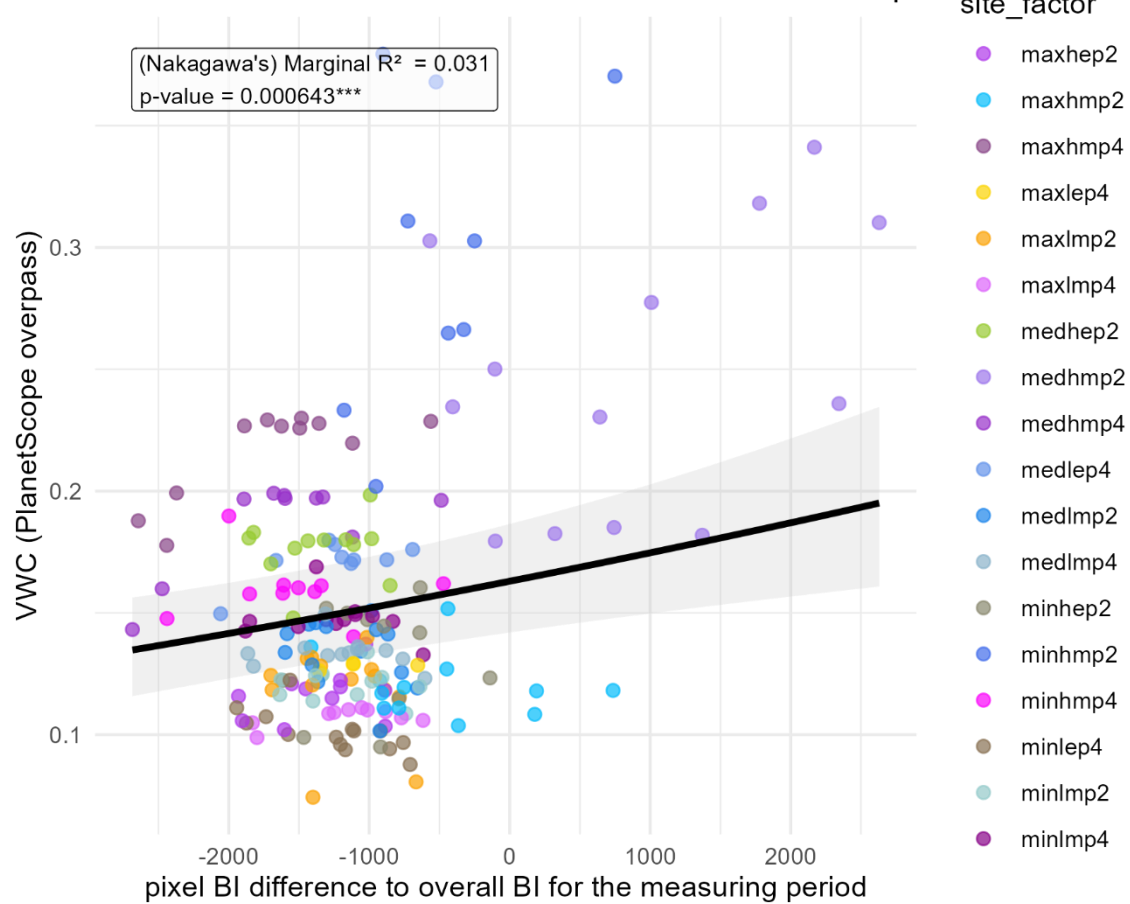
Model with site_id as random effect



BI difference to overall mean BI (not per Pixel)

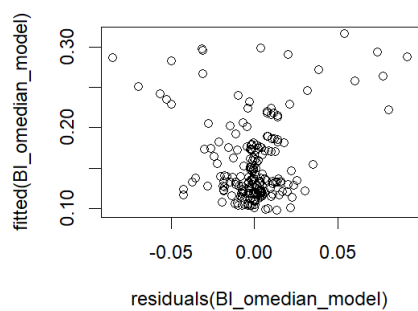


BI to overall mean BI difference–VWC Relationship

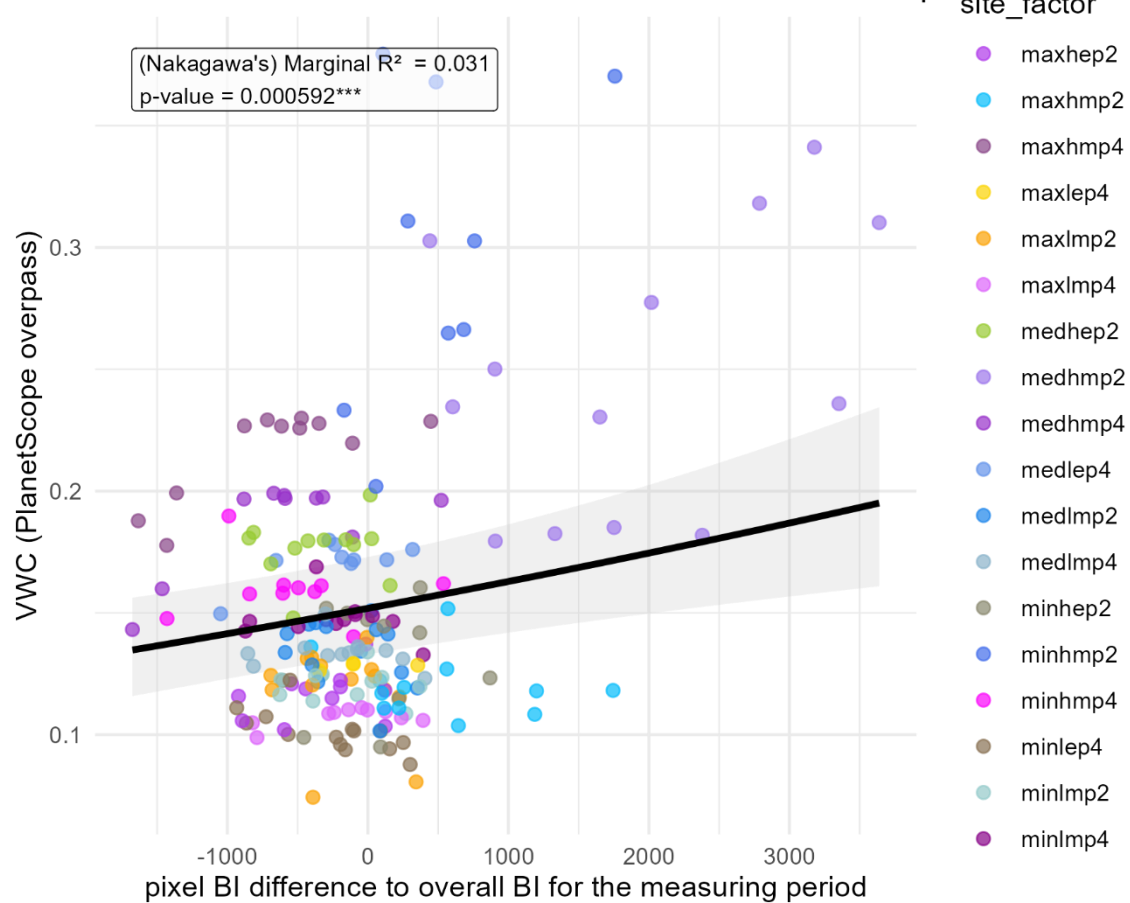


BI difference to overall median BI (not per Pixel)

Not shure if mean or median will be better, I thought that the glaciers may lead to a ,for our purpose, overestimated mean, therefore maybe the median is the better choice?

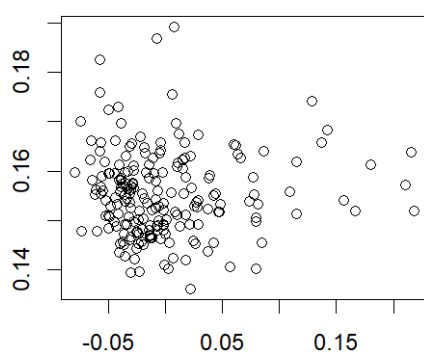


BI to overall median BI difference–VWC Relationship



Seems to make no difference

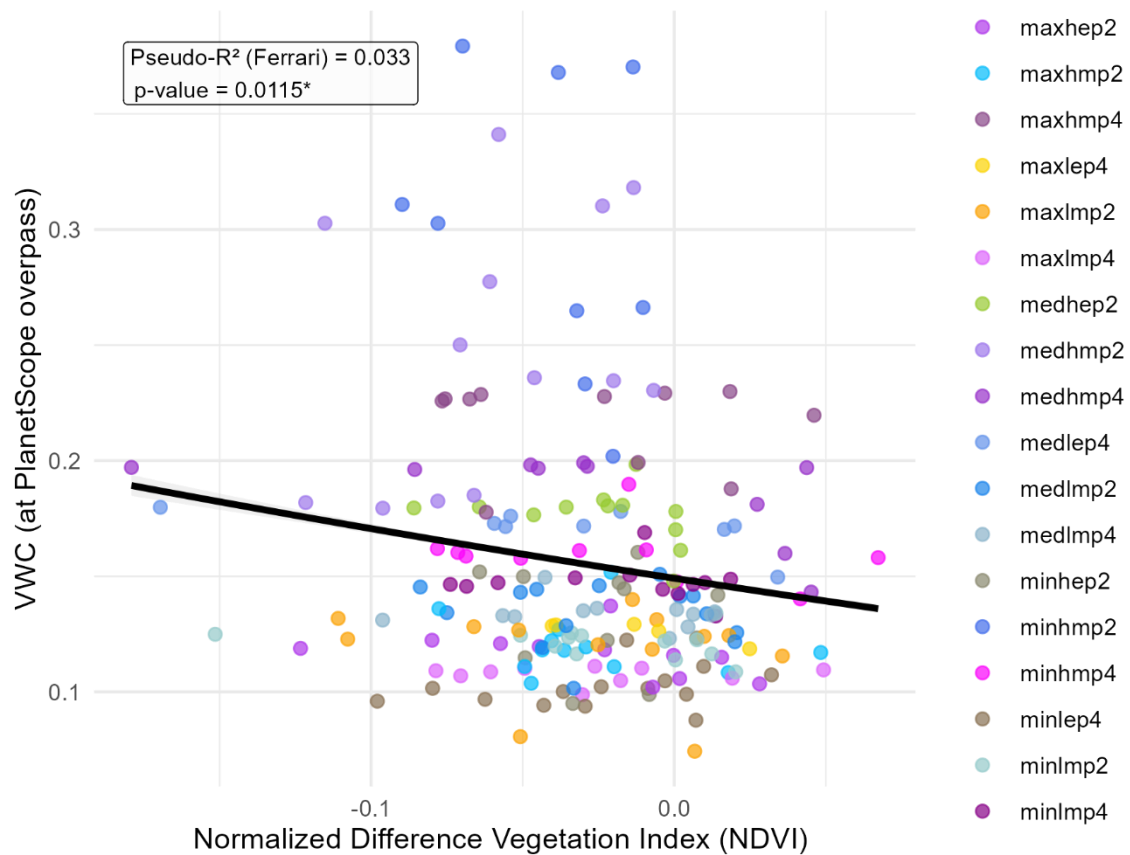
NDVI (as water proxy- but probably better NDWI)



Ok, no obvious pattern

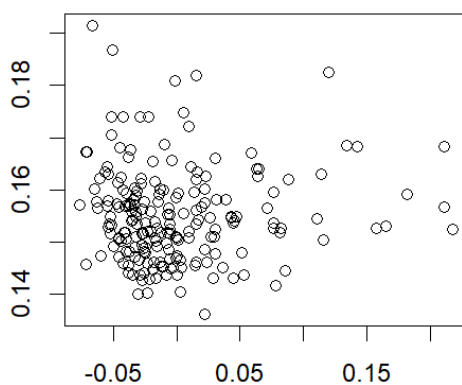
NDVI–VWC Relationship

for all dates where PlanetScope scenes are available



If site_id as random effect is included its not significant anymore and if single dates are looked at not all are significant

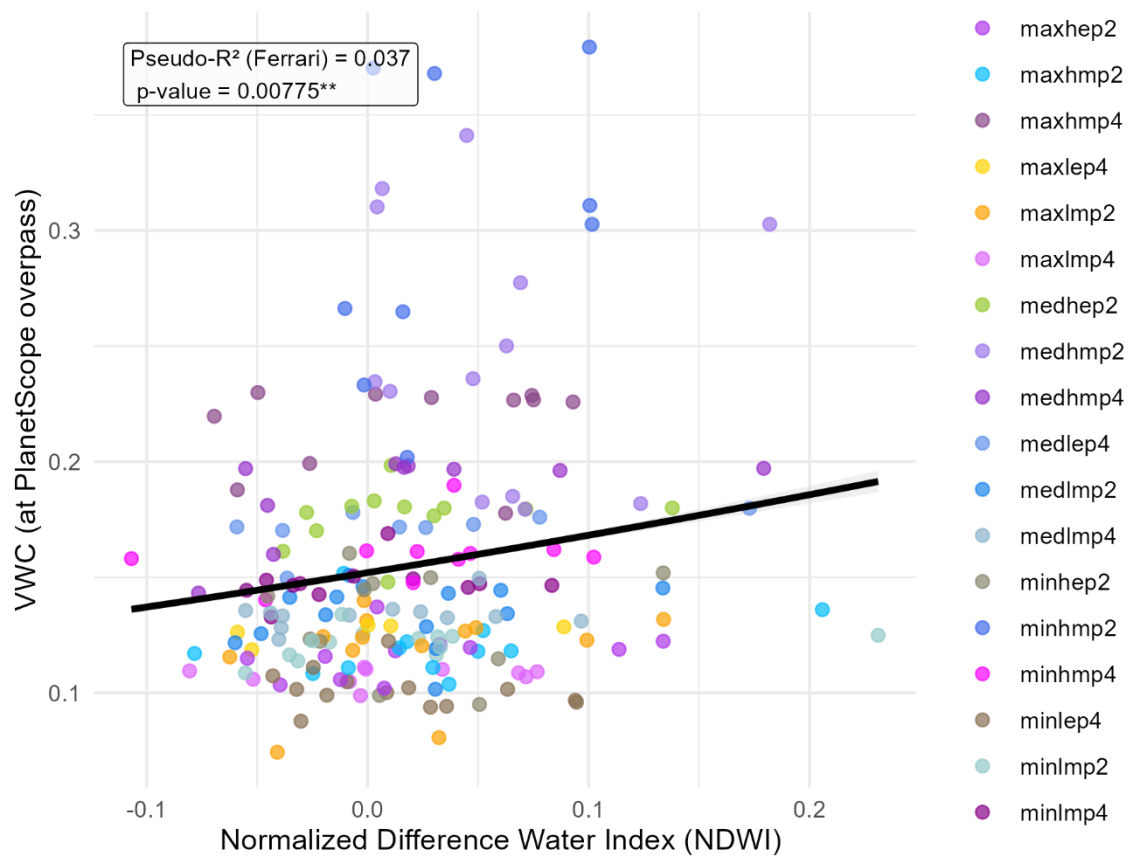
NDWI- VWC relationship



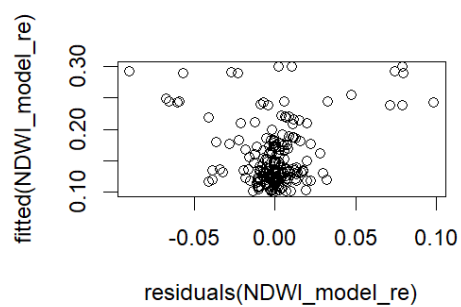
okay, no obvious pattern

NDWI–VWC Relationship

for all dates where PlanetScope scenes are available



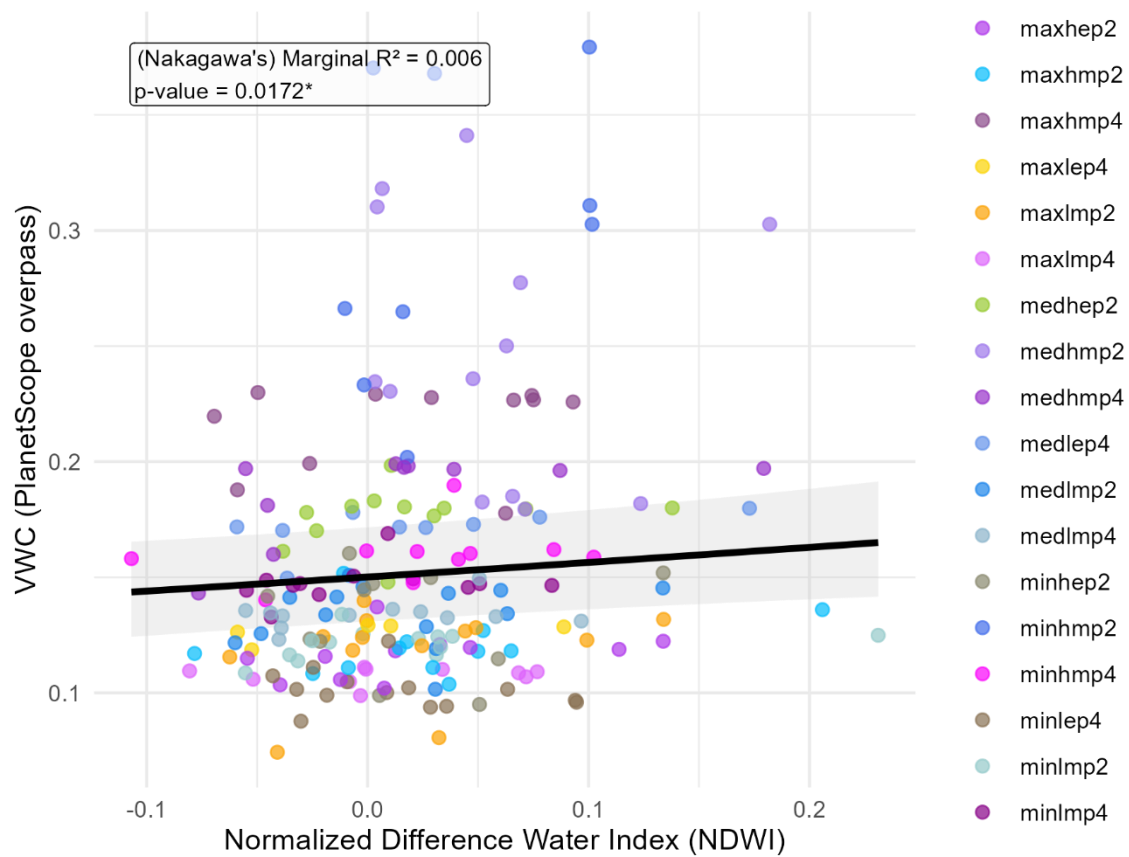
NDWI with site_id as random effect



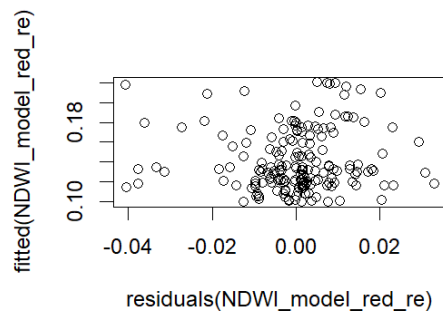
should be okay

NDWI–VWC Relationship (Random Intercept GLMM)

Model: $VWC \sim NDWI + (1 | \text{site})$



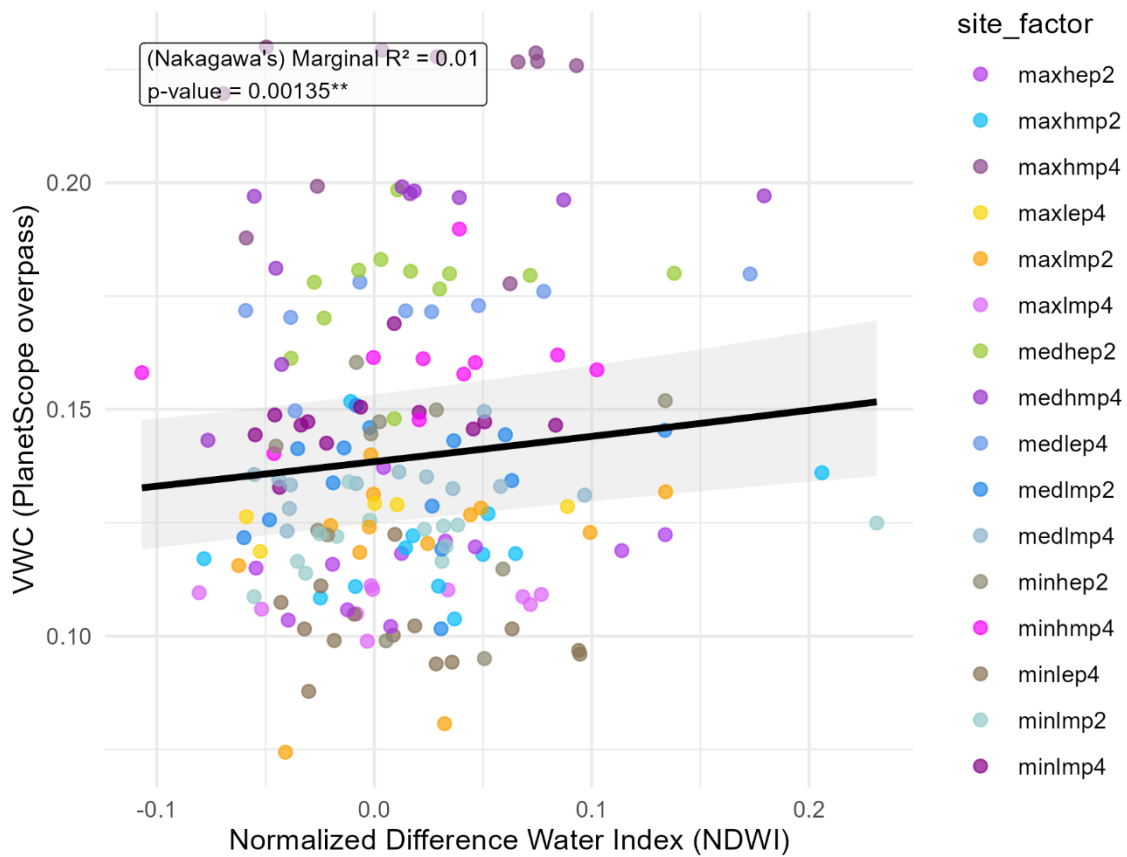
NDWI with random effect and without two wettest spots



looking good

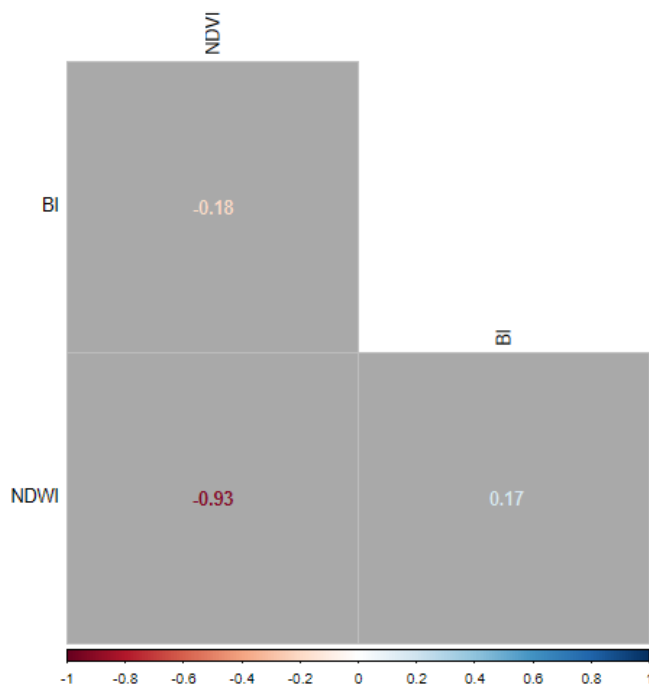
NDWI–VWC Relationship (Random Intercept GLMM)

without sites med-/minhmp2



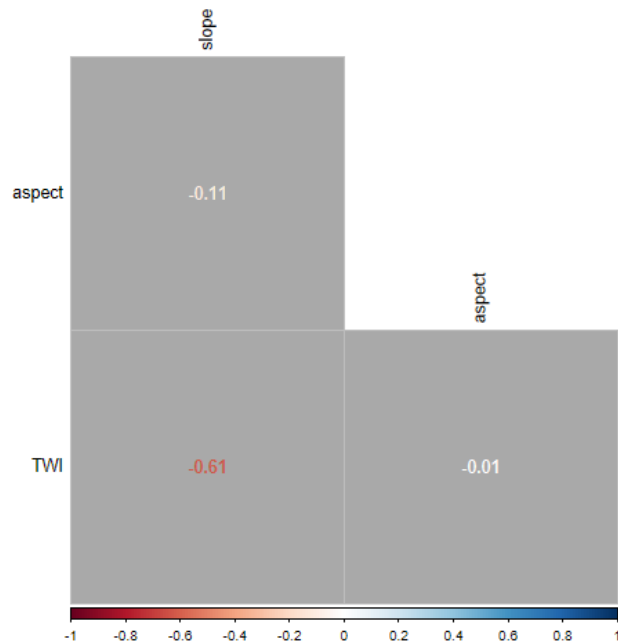
Compared to the model with all sites, this one explains a bit more variance.

Correlations between Multispectral Indices



This correlation matrix indicates, that no correlation problem occurs for BI to NDVI or NDWI, but NDWI and NDVI are co-correlated. Therefore only one of them will be used in the model training.

Because the NDWI fits conceptually better and is higher significant and explains more variance, this index will be used for the model training.



Multicollinearity daily global radiation and daily insolation time:

```
cor(traindat$daily_glob_rad, traindat$daily_insol, use = "complete.obs")  
[1] 0.6830452
```